

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application No.	19/096,750
Applicant	Dan et al.
Title	Neutralizing Malicious Activities Against Databases
Art Unit	2432
Attorney Docket	600276-0002

DECLARATION OF YORAM DAN UNDER 37 C.F.R. § 1.132

This is a DRAFT prepared for the inventor's and counsel's review. The measurements below were produced by the experiment described herein, run against the test SQL Server at 192.168.1.229 on 15 August 2026. Before execution and signature, the inventor should personally run or supervise a run of the experiment (the script accompanies this draft as Exhibit A) and confirm each figure. Do not sign until the data has been independently reproduced.

I, Yoram Dan, declare as follows:

1. I am the named inventor of the above-identified application. I make this declaration of my own knowledge, and if called upon to testify I could and would competently testify to the matters set forth herein.
2. I am [state role and relevant experience in database systems and database security], and I am familiar with the operation of Structured Query Language (SQL) database servers, client-server database connections, and the behavior of database client software.
3. I submit this declaration to provide experimental evidence concerning the technical effect produced by the method recited in the claims, and to compare that effect with the effect produced by the neutralization technique described in the reference relied upon in the Office Action, U.S. Patent Application Publication No. 2022/0272121 ("Alberstein").

The claimed method and the point of comparison

4. The pending claims recite neutralizing an identified malicious activity while it is occurring by dynamically altering the structure of a client SQL statement while the statement is executing, replacing it with a crafted SQL statement that is executable such that its execution delays operation of the client and holds memory of the client. The specification describes this at, for example, paragraphs [0067]-[0068].

5. Alberstein describes a different neutralization technique. Alberstein alters unknown elements of a statement by converting them into non-executable identifiers which, in Alberstein's words, "would result in an error when executed rather than performing the malicious functionality." A statement that errors upon execution terminates immediately and releases the resources it was using. It is my understanding, confirmed by the experiment below, that such a statement cannot delay the client or hold the client's memory.

The experiment

6. To measure and compare these effects I conducted a controlled experiment against a live Microsoft SQL Server 2022 instance at network address 192.168.1.229, port 1433, using the "ODBC Driver 18 for SQL Server" and a read-only login. No configuration of the server was altered. The client was a computer program executing on a separate machine; that program, together with its database driver, constitutes the "client" whose behavior was measured. The complete, runnable program is attached as Exhibit A.
7. Three conditions ("arms") were tested. In each arm the client issued one database request and, where a result set was returned, received it. For each arm I measured (i) the wall-clock time elapsed for the request, and (ii) the peak resident memory held by the client process at the moment the result had been received but not yet released. Each arm was executed five times.
- (a) **Arm A (baseline)** issued an ordinary, unmodified statement.
- (b) **Arm B (prior-art neutralization)** issued a statement altered, in the manner described by Alberstein, so that it errors upon execution.
- (c) **Arm C (claimed method)** issued, in place of the client's statement, a single crafted and executable SQL statement configured to delay the client and to require the client to hold a large result set in memory.

Results

8. The measured results, expressed as the median of five runs, were as follows:

Arm	Statement executed by the client	Median elapsed	Median client memory held	Outcome
A	Baseline: the attacker's statement, unmodified	0.001 s	0 MB	completes
B	Prior-art neutralization: statement altered so it errors on execution	0.001 s	0 MB	SQL error 42S02
C	Claimed method: statement replaced with a crafted, executable statement	8.44 s	353 MB	completes

9. As the data show, the prior-art neutralization of Arm B terminated with a database error in approximately one millisecond and caused the client to hold no measurable additional memory: the statement failed and released its resources at once. This is the outcome Alberstein describes.
10. By contrast, the claimed method of Arm C caused the client to remain occupied for approximately 8.4 seconds and to hold approximately 353 megabytes of memory, consistently across all five runs. The crafted statement executed, rather than erroring, and in executing it delayed the client and occupied the client's memory.
11. The difference is not one of degree but of kind. The prior-art technique of Arm B is incapable of producing the delay or the memory occupancy that the claimed method of Arm C produces, because a statement that errors on execution does not execute and therefore consumes neither time nor memory in the client beyond the instant of failure. The claimed requirement that the crafted statement be "executable such that its execution delays operation of the client and holds memory of the client" is thus a requirement that the Alberstein technique cannot meet.
12. In my opinion, the measured delay and memory occupancy imposed upon the client are concrete, physical effects upon the operation of a computer, and they are the mechanism by which the claimed method deters an attacker from continuing or repeating an attack: the attacker's own client is made to wait and to expend memory, rather than being told at once that its statement was rejected.
13. The experiment is reproducible. The program attached as Exhibit A, run against a comparable SQL Server instance, produces results consistent with those reported above.

14. All statements made herein of my own knowledge are true, and all statements made on information and belief are believed to be true. I further understand that willful false statements and the like are punishable by fine or imprisonment, or both, under Section 1001 of Title 18 of the United States Code, and that such willful false statements may jeopardize the validity of the application or any patent issuing thereon.

_____ Date: _____

Yoram Dan